

Comparison of Incremental Nonlinear Dynamic Inversion Control and Proportional Integral Derivative Control for F-16 Aircraft Attitude Control

Mike Timmerman*

This paper presents a comparison between the Incremental Nonlinear Dynamic Inversion and Proportional Integral Derivative control strategies. Both control laws are tuned to give an equal performance in the nominal case as a basis for further comparison. An assessment is made based on a complex tracking task, changes in dynamic pressure and center of gravity, model mismatch and system faults. Based on this analysis, it is concluded that INDI control performs better in these scenarios than the PID control.

I. Introduction

The aerospace industry commonly uses proven technologies due to the long and expensive procedures of getting new technologies flight certified. Current research is ahead of what is actually implemented in industry. Often, matured and flight proven technology is also preferred as to mitigate risks of new technology failing. Specifically for fixed-wing aircraft, use is made of classical control techniques, such as PID controllers with gain scheduling. However, the introduction of digital flight control systems (FCS) has allowed for more of the advanced flight control systems (AFCS) to be implemented on aircraft which allows for better handling qualities and performance [1].

Of special interest to this paper is the development of nonlinear dynamic inversion (NDI) as flight control in industry. As such, Airbus has implemented an NDI guidance control law for tracking the runway axis during taxiing of a commercial airliner [2]. The German Aerospace Center has implemented dynamic inversion on their Advanced Technologies Testing Aircraft System (ATTAS) [3]. As part of the Vectored Thrust Aircraft Advanced Control (VAAC) program, Nonlinear dynamic inversion was also applied to the VAAC Harrier. Boeing and NASA developed a prototype fighter aircraft: the X-36, a tailless aircraft built to demonstrate feasibility of using new flight control technologies to improve maneuverability and survivability of future fighter aircraft. Its flight control law was based on reconfigurable dynamic inversion [4]. Honeywell has also been a contributor to the development of a dynamic inversion

approach to flight control design. Their approach is MACH (Multi-Application Control), which is a modular, nonlinear multivariable design approach that blends classical control design with inversion.

The advantage of NDI over classical techniques is that NDI avoids gain-scheduling and directly incorporates nonlinearities into the control laws. For this it requires a model of the aircraft dynamics which is then inverted. This however means that potential model mismatches could lead to performance degradation and even unstable situations, which is the major drawback of NDI. In recent years, developments have been made in an extension of NDI, namely incremental nonlinear dynamic inversion (INDI) which decreases the controller's dependency on vehicle model thereby increasing controller robustness.

INDI as a flight control system (FCS) has been applied to various space and aeronautical applications [5–8] and flight tested on a micro aerial vehicle (MAV) [9, 10]. This research will further investigate the performance and robustness characteristics of INDI applied as the F-16 attitude control as compared to classical PID control.

The paper is structured as follows: first, a detailed description of the system which is to be controlled, the F-16 aircraft, is given in [section II](#). Then, a theoretic description of the control techniques is given in [section III](#). In [section IV](#), the control laws are derived for the specific system and subsequently tuned. [section V](#) describes the methodology used to perform the analysis. Finally, the results of the analysis are presented in [section VI](#).

*Student, TU Delft Faculty of Aerospace Engineering, Kluyverweg 1, 2629HS Delft

II. Description of the System

The General Dynamics F-16 Fighting Falcon is a 4th generation single-engine multirole fighter aircraft originally developed by General Dynamics for the United States Air Force. Over 4,600 aircraft have been built since production was approved in 1976.

A. Aircraft Configuration

The F-16 utilises relaxed static stability/fly-by-wire flight control system which helps to make the aircraft agile. Relaxed stability means that the aircraft has the tendency to change its pitch and bank angle spontaneously. Furthermore, when disturbed in pitch or roll, the aircraft will continue to pitch or roll in the direction of the disturbance. This is complemented with fly-by-wire system which replaces conventional manual flight controls with an electronic interface.

Typically, aircraft designed to reach beyond transonic speeds are designed with a high sweep angle. The sweep angle acts to postpone the transonic drag rise since the component of the airflow perpendicular to the leading edge is reduced. Furthermore, the requirement for low supersonic wave drag and high maneuverability leads to high-speed fighter aircraft to be designed with a low aspect ratio.

The F-16 itself has a cropped-delta wing with a leading-edge sweep of 40° and a straight trailing edge and is designed to have an aspect ratio of 3. The stubby wings allow the aircraft structure to withstand very high forces during maneuvers. Furthermore, they also reduce moment of inertia about the longitudinal axis and aerodynamic damping moments during rolling, hence the configuration stimulates a high roll acceleration and high maximum roll rate.

B. Longitudinal Characteristics

The F-16 aircraft has undergone many tests to identify its performance and stability characteristics. Some of the main longitudinal characteristics are highlighted in this part, as also described in [11].

Starting off, from wind-tunnel flow visualization tests, it has been determined that the the outer wing panels stall near $\alpha = 20^\circ$. However, the entire aircraft keeps producing lift up to a maximum C_L value at $\alpha = 35^\circ$.

Furthermore, the aircraft configuration has a slight level of static pitch instability at nominal center-of-gravity position at low speeds. Therefore, pitch stability has to be ensured artificially.

Another important characteristic for longitudinal stability at high angles of attack is that the horizontal stabilizers lose effectiveness in providing a nose-down

moment due to the surfaces stalling for angles of attack greater than 25° .

Lastly, a sharp loss in stabilator effectiveness for sideslip angles greater in magnitude than 10° is observed.

Hence, in order to retain control effectiveness of the horizontal stabilizer the angle of attack should be limited to below 25° and sideslip angles to at least below 10° .

C. Lateral-Directional Characteristics

The configuration is found to be both directionally and laterally statically stable for angles of attack up to 28° .

The rudder effectiveness is high and constant over the entire range of operational angles of attack. The roll effectiveness of the ailerons is also high over the angle of attack operational range, while at angles of attack above 20° , adverse yaw is produced by the ailerons. However, the adverse yaw remains small compared to the moments produced by the rudder up to an angle of attack of 40° .

An evaluation of the dynamic lateral-directional behavior of the aircraft configuration shows that the dutch roll, spiral and roll modes of motion are all damped and stable modes for values up to $\alpha = 30^\circ$.

D. Characteristic Data

The main geometric, mass and aerodynamic characteristic data of the F-16 Fighter Jet model are given in Table 1. This data is taken from [11, pg. 43].

Table 1: Characteristic data for the F-16 fighter jet

Geometric Characteristics	
Wing Area S [m^2]	27.87
Wing Span b [m]	9.144
Mean Aerodynamic Chord $\bar{c}$ [m]	3.45
Mass characteristics	
Mass m [kg]	9298.64
Center of Gravity Location	0.35 $\bar{c}$
Roll Moment of Inertia J_{xx} [kgm^2]	12,874.8
Pitch Moment of Inertia J_{yy} [kgm^2]	75,673.6
Yaw Moment of Inertia J_{zz} [kgm^2]	85,552.1
Product Moment of Inertia J_{xz} [kgm^2]	1331.4
Product Moment of Inertia J_{xy} [kgm^2]	0.0
Product Moment of Inertia J_{yz} [kgm^2]	0.0
Dimensionless coefficients	
$C_{l_{\delta_a}} = -0.0530$	$C_{l_{\delta_r}} = 0.0140$
$C_{n_{\delta_a}} = -0.0170$	$C_{n_{\delta_r}} = -0.0520$
$C_{m_{\delta_e}} = -0.0102$	

E. Control surfaces description

A typical fixed-wing aircraft consists of three main control surfaces, namely the elevator, ailerons and rudder, respectively controlling the pitch, roll and yaw angle of the aircraft. This is done by deflecting the control surface through a certain angle, creating a moment. The control surfaces have a maximum angle through which they can be deflected. Furthermore, they also have a maximum deflection rate and a time-lag. The characteristics of the control surfaces for the F-16 aircraft are shown in Table 2. Additionally, the F-16 has leading edge flaps (LEF), but these are not modelled in the simulation.

Table 2: Actuator data for the F-16 aircraft

Symbol	Control Surface Name	Deflection Limit	Rate Limit	Time Constant
δ_e	Elevator	$\pm 25.0^\circ$	$60^\circ/s$	$0.0495s$
δ_a	Ailerons	$\pm 21.5^\circ$	$80^\circ/s$	$0.0495s$
δ_r	Rudder	$\pm 30.0^\circ$	$120^\circ/s$	$0.0495s$

The moments lead to a change in the rotational motion of the aircraft according to Equation 4, hence the moment vector acting on the aircraft due to a control surface deflection is an input to change the aircraft's rotational motion, with the angular velocity vector being the state. A relation between the deflections of the control surfaces and the applied moment is given in Equation 1, assuming that the control moments depend linearly on the control surface deflections.

$$\mathbf{M}_c = \frac{1}{2} \rho V^2 S \begin{bmatrix} bC_{l_{\delta_a}} & 0 & bC_{l_{\delta_r}} \\ 0 & \bar{c}C_{m_{\delta_e}} & 0 \\ bC_{n_{\delta_a}} & 0 & bC_{n_{\delta_r}} \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_e \\ \delta_r \end{bmatrix} \quad (1)$$

F. Attitude kinematics

In order to control the attitude of an aircraft, the relation between the attitude angles and angular rates are required, which is based on kinematic equations using Euler angles and independent of aircraft characteristics, given in Equation 2, [12, pg. 20].

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi \tan\theta & \cos\phi \tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi \sec\theta & \cos\phi \sec\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (2)$$

where ϕ , θ and ψ are the roll, pitch and yaw angles respectively and p , q and r are the roll pitch and yaw rates respectively in the body-fixed reference frame. These are depicted in Figure 1.

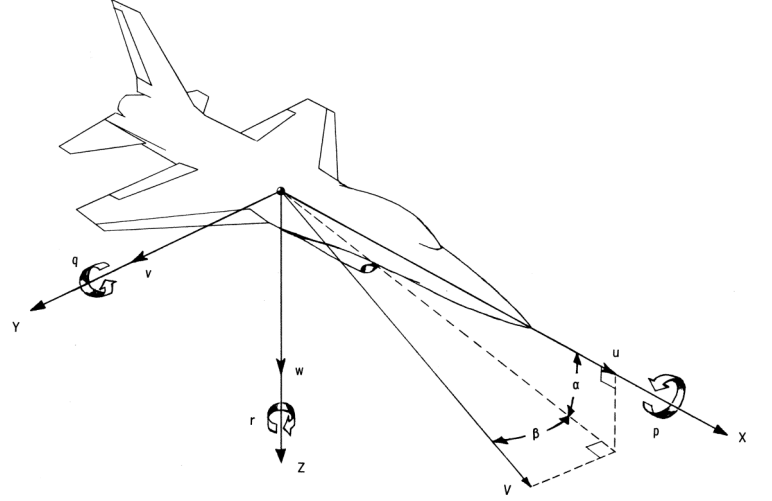


Figure 1: Body-fixed reference frame with a forward-right-down coordinate system. Roll rate q , pitch rate p , yaw rate r , velocity components u , v , w and angle of attack α , sideslip angle β indicated.

G. Attitude dynamics

The attitude dynamics of a fixed-wing aircraft, in this case the F-16 aircraft, involves the description of the rotational motion of the rigid body under the influence of external forces. Consider Euler's rotational equation of motion in vector form [12, pg. 36]:

$$\mathbf{M}^{bf} = J^{bf} \dot{\boldsymbol{\omega}}_{b/i}^{bf} + \boldsymbol{\omega}_{b/i}^{bf} \times J^{bf} \boldsymbol{\omega}_{b/i}^{bf} \quad (3)$$

where $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity vector, $\mathbf{M} \in \mathbb{R}^3$ is the external moment vector, and J is the inertia matrix of the rigid body. The superscript bf indicates that the variable is w.r.t. the body frame, and the subscript b/i indicates that the angular velocity describes the rotation of the body frame w.r.t. the inertial frame. For the purpose of control system design, the earth-fixed reference frame is assumed to be the inertial reference frame.

Rewriting the equations of rotational motion given in Equation 3 to give the time rate of change in angular velocity gives the following set of differential equations:

$$\dot{\boldsymbol{\omega}}_{b/i}^{bf} = (J^{bf})^{-1} [\mathbf{M}_c^{bf} + S(\boldsymbol{\omega}_{b/i}^{bf}) J^{bf} \boldsymbol{\omega}_{b/i}^{bf}] \quad (4)$$

where:

$$\boldsymbol{\omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad J = \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{xy} & J_{yy} & J_{yz} \\ J_{xz} & J_{yz} & J_{zz} \end{bmatrix} \quad \mathbf{M} = \begin{bmatrix} L \\ M \\ N \end{bmatrix} \quad (5)$$

$$S(\boldsymbol{\omega}) = \begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix} \quad (6)$$

III. Control Laws

When considering a nonlinear system, linear control techniques can be applied through linearizing its system dynamics. However, for some nonlinear systems, these linear control techniques only perform well in a small area around the linearization state. In order to control such systems effectively, a nonlinear control technique is required which is robust, making it able to account for model mismatches.

PID control is model-free and process-based. The controller parameters can be tuned by trial and error which makes it an attractive option for systems for which it is hard to obtain an accurate model.

Incremental nonlinear dynamic inversion (INDI) is an extension to the nonlinear control technique, nonlinear dynamic inversion (NDI). The concept of these techniques is to cancel the nonlinearities so that the closed-loop dynamics is in a linear form, allowing conventional linear control to be applied.

A. NDI control

A control law designed using nonlinear dynamic inversion (NDI) relies on an accurate description of the system as it is assumed that the model used in the control law behaves exactly as the actual system. Furthermore, NDI relies on complete and accurate knowledge about the state of the system. The control law is derived starting from the general nonlinear dynamics, affine in control input, given in Equation 7.

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u} \quad (7)$$

The control law is then obtained by inverting the dynamics in Equation 7, given that the matrix $\mathbf{G}$ is invertible for the domain of interest of $\mathbf{x}$. The result is given in Equation 8.

$$\mathbf{u} = \mathbf{G}^{-1}(\mathbf{x}) [\boldsymbol{\nu} - \mathbf{f}(\mathbf{x})] \quad (8)$$

A linear input-output relation is obtained between the state and the virtual input $\boldsymbol{\nu}$, canceling the nonlinearities:

$$\dot{\mathbf{x}} = \boldsymbol{\nu} \quad (9)$$

In general, the introduced virtual input $\boldsymbol{\nu}$ can be designed to solve the problem of output tracking.

B. INDI control

In INDI-based control laws, the system is expressed in an incremental form in order to improve robustness of the closed-loop system by reducing its dependency on the model and exact knowledge on the plant dynamics. The incremental form considers the influence of increments of control commands in the dynamics.

The control law is derived from the general non-linear dynamics given in Equation 10.

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \quad (10)$$

The system of Equation 10 can be linearized about the current state given by the latest available measurements. The linearization, which is given in Equation 11, is based on assumptions of a small sampling time and instantaneous control effectors.

$$\begin{aligned} \dot{\mathbf{x}} \approx \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) &+ \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} (\mathbf{x} - \mathbf{x}_0) \\ &+ \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} (\mathbf{u} - \mathbf{u}_0) \end{aligned} \quad (11)$$

Based on the time-scale separation principle, it is assumed that the control effectors are sufficiently fast compared to the system state. Hence, the change in input, $\mathbf{u}$, is significantly faster than the change in system state, $\mathbf{x}$. It can thus be assumed that $\frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} (\mathbf{x} - \mathbf{x}_0) \ll \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} (\mathbf{u} - \mathbf{u}_0)$. This leads to a simplification of Equation 11 given in Equation 12.

$$\dot{\mathbf{x}} = \dot{\mathbf{x}}_0 + \mathbf{G}(\mathbf{x}_0, \mathbf{u}_0)(\mathbf{u} - \mathbf{u}_0) \quad (12)$$

Equation 12 is used to obtain the control law by defining the virtual input $\boldsymbol{\nu} = \dot{\mathbf{x}}$. The physical control input $\mathbf{u}$ can be computed using Equation 13 using measurements $(\dot{\mathbf{x}}_0, \mathbf{x}_0, \mathbf{u}_0)$ and the virtual control input $\boldsymbol{\nu}$, which is to be designed.

$$\mathbf{u} = \mathbf{u}_0 + \mathbf{G}^{-1}(\mathbf{x}_0, \mathbf{u}_0)(\boldsymbol{\nu} - \dot{\mathbf{x}}_0) \quad (13)$$

This control law is only dependent on the control effectiveness $\mathbf{G}$, as well as on the sensor measurements. Hence, dependence on system knowledge of the plant is reduced. However, making it more susceptible to measurement noise, bias, misalignments, etc.

C. PID control

PID controllers consist of three components; proportional, integral and derivative controls. The proportional term includes proportional changes of the error to the control output, integral term sums the error over time and corrects the output by reducing steady-state errors, derivative term tracks the rate of change of the error and thereby accounts for unusual variations. The control law based on PID control is given in Equation 14.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (14)$$

where $e(t)$ is the difference between the reference signal and the measured system output and (K_p, K_i, K_d) are the gains to be tuned.

IV. Attitude Control Law Design

This section derives the control laws of the two control techniques for the specific system under consideration. These are subsequently tuned.

A. INDI controller

The INDI controller consists of two cascaded control loops. The inner loop controls the angular velocity using INDI, while the outer loop is constructed to control the attitude using NDI. It is structured this way because the outer loop does not require system knowledge, hence NDI can be utilised. This inner loop on the other hand does require system knowledge, hence INDI is applied.

Angular Velocity Inner Loop

The inner loop is designed to track a reference angular velocity vector ω_{ref} . To obtain an incremental form of Euler's rotational equations, as in Equation 11, consider a Taylor series expansion of Equation 4 to obtain a first-order approximation of $\dot{\omega}$

$$\begin{aligned} \dot{\omega} \approx & J^{-1}(S(\omega_0)J\omega_0 + M_{c_0} + M_{a_0} + M_{d_0}) \\ & + \frac{\partial}{\partial \omega} [J^{-1}(S(\omega)J\omega + M_a)]_{\omega=\omega_0} (\omega - \omega_0) \\ & + \frac{\partial}{\partial \delta} [J^{-1}M_c]_{\delta=\delta_0} (\delta - \delta_0) \end{aligned} \quad (15)$$

Consider δ_0 and ω_0 to be control and state an incremental instance in time before δ and ω . Then the first term is the angular acceleration of the previous time step $\dot{\omega}_0$. Furthermore, considering the time-scale separation principle. the second term, partial to ω , is much smaller than the third term, partial to δ . It is assumed that demanded deflection is achieved instantaneously. Also considering the linear relation between deflection and control moment, Equation 1:

$$\dot{\omega} = \dot{\omega}_0 + G(\delta - \delta_0) \quad (16)$$

where G is given by Equation 18. Since angular acceleration is not readily available from measurements, backward Euler finite-difference scheme is used to approximate it using the angular velocity, Equation 17.

$$\dot{\omega}_0 = \frac{\omega_0 - \omega_{-1}}{\Delta t} \quad (17)$$

$$G = J^{-1} \frac{1}{2} \rho V^2 S \begin{bmatrix} bC_{l_{\delta_a}} & 0 & bC_{l_{\delta_r}} \\ 0 & \bar{c}C_{m_{\delta_e}} & 0 \\ bC_{n_{\delta_a}} & 0 & bC_{n_{\delta_r}} \end{bmatrix} \quad (18)$$

Inverting Equation 16 gives the control law intended to control the angular velocity, Equation 19, where it is assumed that $u = \delta$

$$u = u_0 + G^{-1}(\nu_{in} - \dot{\omega}_0) \quad (19)$$

The system $\dot{x} = \dot{\omega} = \nu_{in}$ is designed based on a PID-control structure for the virtual control law.

Attitude Outer Loop

The attitude outer loop is constructed to control the attitude angles. The outer loop is nested around the inner loop, where the reference commands for the inner loop are provided by the outer-loop control. The time-scale separation principle then says that it can be assumed that the angular velocities of the inner loop reach steady-state instantaneously, i.e. the inner loop dynamics can be neglected.

The relation between the angular velocities, which are the inputs to the inner loop, and the attitude angles, which are to be controlled, is given by the kinematic equations given in Equation 2. The equations are independent of any aircraft characteristics, hence NDI is used for the outer loop. Then, using NDI as control law, specified in Equation 8, the outer loop control law is:

$$u_k = \omega = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\phi) & \cos(\theta)\sin(\phi) \\ 0 & -\sin(\theta) & \cos(\theta)\cos(\phi) \end{bmatrix} (\nu_{out}) \quad (20)$$

The system $\dot{x} = \nu_{out}$, with $x = \eta = [\phi, \theta, \psi]$, is a first-order virtual control law, for which the control is designed based on PID-control. The block diagram visualizing this control loop and its interaction with the inner control loop is given in Figure 2.

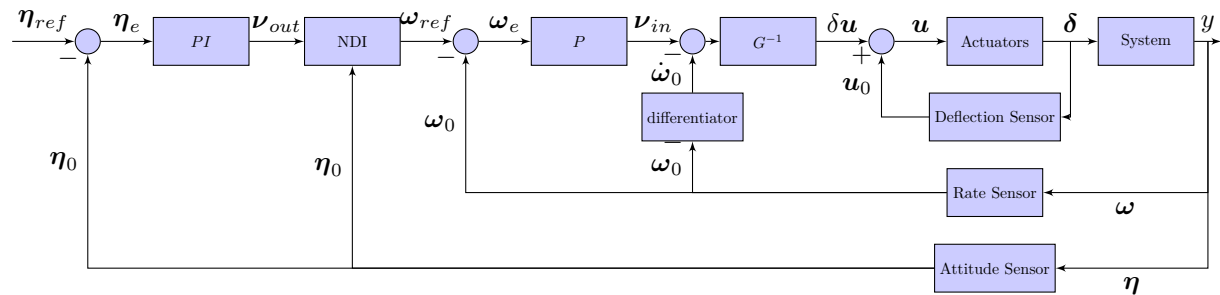


Figure 2: Block diagram of INDI controller

B. PID Controller

The attitude controller using PID control is based on successive loop closure. The loops are nested by arranging that the reference commands for the inner loop are provided by the outer-loop controller. In this case, the outer attitude control loop provides a reference for the inner angular velocity control loop. This cascading control loop architecture is commonly used to control the attitude dynamics of a system, where it is assumed that the inner loop has faster dynamics than the outer loop. In that case, it can be assumed that the inner loop dynamics reach the steady-state instantaneously. This is the same principle as given in the description of the INDI control technique in [subsection A](#), namely the time-scale separation principle.

The controller consists of three separate control channels. One to control each attitude angle; roll, pitch, yaw. Each channel consists of its own PID gains, hence three separate control inputs are obtained; ailerons, elevator, rudder. The control loop for each channel has the architecture given in [Figure 3](#).

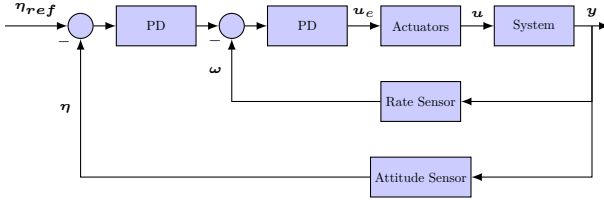


Figure 3: Block diagram of PID controller

C. Controller Tuning

In order to compare both controllers, the tuning is to be done in such a way that there is no bias in their performance, i.e. one controller cannot be tuned worse than the other leading to an unfair comparison. This is accomplished through tuning the gains of each controller based on a certain cost function. The cost function minimizes the sum of the root mean squared error of each attitude trajectory. To ensure that the controllers are not tuned to be too aggressive, which leads to large amounts of overshoot and oscillations, the cost function also includes a penalty in case of an overshoot of more than 15%, ν . The performance index is given in [Equation 21](#).

$$V(\boldsymbol{\eta}) = \sum_{k=1}^3 \sqrt{\frac{1}{N} \sum_{i=0}^N ((\eta_{k_i})_{ref} - \eta_{k_i})^2 + \nu} \quad (21)$$

where η_{k_i} is the $k \in \{1, 2, 3\}$ attitude angle at given time step $i \in \mathbf{N}$. $(\eta_{k_i})_{ref}$ is the attitude reference

vector at given time step k . $t_f = N\Delta t = 15[\text{sec}]$ is the final time of the simulation. The function $V(\boldsymbol{\eta})$ is minimized using the build in Matlab search function "*fminsearch()*" at specific trim conditions and for a particular reference. The function finds the minimum of an unconstrained multivariable function using the simplex search method of Lagarias.

The trim conditions are chosen to be in the center of the flight envelope, so the aircraft is trimmed at a velocity of $600[\text{ft/s}]$ and altitude of $25000[\text{ft}]$. To obtain the gains of one of the control loops, the gains are varied according to minimizing the performance index, given a doublet as reference trajectory while the other control loops are set to track the trim states. For the roll and pitch control loops, magnitude of the doublet is $30[\text{deg}]$, while for the yaw control loop, magnitude of $5[\text{deg}]$ is used. Hence, to tune the gains of the entire controller, three optimization cycles are completed; one for each control channel.

The results for both the PID- and INDI controller gains are given in [Table 3](#) and [Table 4](#), respectively. The initial gains are obtained by manually setting them such that the PID and INDI controller have a similar response. Since a steady-state offset is present for the PID controller on the yaw angle channel, an integral gain is added to the inner yaw control loop with a value of -0.6 to ensure the proper tracking of the reference signal.

Table 3: PID control gains

Loop	PID				Cost $V(\eta)$
	Inner		Outer		
	K_P	K_D	K_P	K_D	
Initial gains	-2.000	-0.300	6.000	1.000	
Roll, $p - \phi$	-2.329	-0.211	6.725	1.012	12.31
Pitch, $q - \theta$	-2.046	-0.290	6.138	1.025	17.09
Yaw, $r - \psi$	-2.240	-0.196	7.351	0.860	2.46
Max. function evals.	5×10^1				
Max. iterations	1×10^1				

Table 4: INDI control gains

Loop	INDI				Cost $V(\eta)$
	Inner		Outer		
	K_P	K_D	K_P	K_D	
Initial gains	5	1	6	1.5	
Roll, $p - \phi$	5.80	0.499	8.143	1.247	11.59
Pitch, $q - \theta$	4.003	0.706	6.239	1.821	22.47
Yaw, $r - \psi$	3.698	0.782	6.801	1.305	2.71
Max. function evals.	5×10^1				
Max. iterations	1×10^1				

V. Methodology

A. Nominal Control System

The nominal controller performance is used as a baseline for further analysis of the controller's performance when tested against certain defects or other effects which might cause controller performance degradation.

The nominal performance is evaluated by giving a reference input for each attitude angle separately. Standard step input characteristics are then derived, such as rise time, transient time, settling time, overshoot, peak, and peak time. These characteristics are compared between INDI and PID control, and should be similar as to have a similar controller performance as a starting point for further comparison.

Also the response to the reference inputs used for tuning is assessed, since no bias in the tuning process should occur, towards tuning one controller better than the other, again a similar response of both controllers is expected.

B. Effect of Center of Gravity Shift

As mentioned in a previous section, the F-16 aircraft is designed with relaxed static stability. Hence, its center of gravity lies behind the neutral point, causing longitudinal static instability. However, while tuning, the center of gravity was kept at $0.29\bar{c}$, giving a static margin of 0.02, which is a statically stable configuration. To evaluate the control laws under the conditions in which the aircraft is intended to operate, an evaluation is made of the control law performance when shifting the center of gravity aft.

The first center of gravity configuration which is evaluated is a position of $0.35\bar{c}$, which is the aircraft's designed nominal center of gravity position. this given the aircraft a static margin of -0.04 .

Secondly, a center of gravity position of $0.41\bar{c}$, which, although outside of the operational center of gravity range of the aircraft, is chosen as to give an indication how severely it affects controller performance, and to see if one controller can still outperform the other. This center of gravity position leads to a static margin of -0.10 .

Both center of gravity positions will be evaluated based on a roll angle and pitch angle reference, and are compared to the response of a stable center of gravity configuration, i.e. center of gravity at $0.29\bar{c}$. Both references are a doublet with a magnitude of $20[deg]$ and step duration of $5[sec]$. Evaluation is also made on the control activity under each condition.

C. Effect of Model Mismatch

A benefit of using incremental non-linear dynamic inversion is its reduced dependency on exact knowledge of the system dynamics. However, despite this, knowledge of the aircraft actuators is still required in the form of the control effectiveness. Therefore, an evaluation is made on controller performance with changes in the control effectiveness term. This can either be due to model mismatch or due to in-flight events. This section, an evaluation is made on the effects of model mismatch, while the next section handles operational defects. As previously shown, control effectiveness depends on the moments of inertia as well as on the control derivatives.

Starting with the control derivatives, to apply uncertainty to the control derivatives, a similar approach is taken as in [13]. The control derivatives are modelled as given in Equation 22, where $C_*(\Delta)$ represents one of the aerodynamic coefficients and $\hat{C}_*(\Delta)$ represents the coefficient subjected to aerodynamic uncertainty. Uncertainty in the coefficients is introduced by the Standard Normal Distribution $\mathcal{N}(0, 1)$. The standard deviation is scaled through the scaling factor n . Evaluation will be made for the following scaling factors $n = [0.05 \quad 0.15 \quad 0.25]$, hence standard deviation ranges between 5% and 25% and mean 0.

$$\hat{C}_*(\Delta) = C_*(\Delta)(1 + n\mathcal{N}(0, 1)) \quad (22)$$

Then the uncertainty in the inertia matrix. Since the aircraft has one symmetry plane, its main inertia terms are along the principal axes of the body-fixed reference frame. Uncertainty in each of these terms will be evaluated according to Equation 23, where J_{n_i} is a nominal inertia term, $\hat{J}_i$ is an inertia term subject to uncertainty, and n is, again, a scaling factor for the Standard Normal Distribution. Since the inertia terms can be determined more accurately, evaluation is made for following scaling factors $n = [0.05 \quad 0.10]$.

$$\hat{J}_i = J_{n_i}(1 + n\mathcal{N}(0, 1)) \quad (23)$$

To analyse the robustness of the control laws against aerodynamic uncertainty and inertial uncertainty, the same reference trajectory is used for the attitude angles. The Root-Mean-Squared Error (RMSE) is determined according to Equation 24, where N represents the total number of time steps, x_{ref} represents the reference state and x is the actual state.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=0}^N (x_{ref} - x_i)^2} \quad (24)$$

D. Effects of System Faults

An aspect of a robust controller is that it is expected to perform well under the assumption of worst-case conditions. Therefore, the performance degradation or possible loss of stability of the controllers designed in this paper are tested against system faults. After reviewing relevant fault-tolerant control literature, the following system faults are identified for further analysis,

1. Sudden change in the moments of inertia terms.
2. Changes in the aerodynamic parameters, i.e. the lift and drag coefficients, due to, for example, icing which increases skin friction drag and reduces the maximum lift attainable.
3. Reduction in control surface effectiveness. For the F-16, the control surfaces are the elevator, rudder and ailerons. This can be caused due to, for example, mechanical friction or damage.
4. Reduced range in maximum deflection of the actuators. This may be caused by actuator failure or damage.

1. Damage to the wings

During operation, the aircraft can get damaged which may lead to a change in moments of inertia during flight due to partial loss of structural elements. Furthermore, a combat aircraft can also be equipped with stores on its wings. While during operations, these stores can be released, leading to changed inertia values.

To analyze these effects on the controller performance, a decrease of 25% in the roll moments of inertia J_{xx} is considered to resemble the usage of stores and loss of wing elements along with a 70% decrease in C_{l_p} since a damaged wing provides decreased roll damping. Furthermore, C_{l_r} is also decreased by 70%.

Along with the changes in the roll moments of inertia and aerodynamic coefficients, two defects of the aileron are considered. Firstly, a reduction in control effectiveness which takes affect through the control derivatives for the aileron $C_{l_{\delta_a}}$ and $C_{n_{\delta_a}}$. This reduction in effectiveness causes a given deflection of the control surface to produce less aerodynamic moment. Since INDI relies on knowledge of the actuators, a sudden change in the control derivatives potentially affects the performance. Therefore, a decrease of up to 50% in aileron control effectiveness is investigated. Secondly, a reduction in maximum range of aileron deflection is investigated. In this case, a new range of $[-5.0^\circ, 5.0^\circ]$ is analysed.

2. Damage to the horizontal tail

In the case of damage to the horizontal tail, the pitch damping as well as longitudinal stability are reduced. Both the stability derivatives C_{m_q} and longitudinal stability coefficient C_{m_α} are reduced by 70%.

Similarly to the damaged wings with defect ailerons, the two defects of the elevator are analyzed. To consider decreased elevator effectiveness, the control derivative $C_{m_{\delta_e}}$ is decreased by 50 %. Furthermore, the maximum range of elevator deflection is set to $[-10.0^\circ, 10.0^\circ]$ in absolute values.

3. Damage to the vertical tail

As a last damage case, damage to the vertical tail is considered. The aerodynamic stability coefficients which are affected are the weathervane stability C_{n_β} and the yaw damping C_{n_r} . To model the effects of a damaged vertical tail, these coefficients are decreased by 70%.

Similarly to the previous two damage cases, two defects of the rudder are analysed. To consider decreased rudder effectiveness, the control derivatives $C_{n_{\delta_r}}$ and $C_{l_{\delta_r}}$ are decreased by 50 % and a reduced maximum range of rudder deflection of $[-10.0^\circ, 10.0^\circ]$ is considered.

A table summarizing all system faults for each specific damage case is shown in [Table 5](#).

Table 5: Overview of system faults

Faults	Damage to wings	Damage to horizontal tail	Damage to vertical tail
Inertia parameters	J_{xx}	-	-
Aerodynamics coefficient	C_{l_p} C_{l_r}	C_{m_q} C_{m_α}	C_{n_β} C_{n_r}
Affected control surface	Ailerons	Elevator	Rudder
Reduced control effectiveness	$C_{l_{\delta_a}}$ $C_{n_{\delta_a}}$	$C_{m_{\delta_e}}$	$C_{n_{\delta_r}}$ $C_{l_{\delta_r}}$
Reduced max. deflection	$[-5.0^\circ, 5.0^\circ]$	$[-10.0^\circ, 10.0^\circ]$	$[-10.0^\circ, 10.0^\circ]$

VI. Results & Discussion

A. Analysis Nominal Behaviour

The analysis of the nominal behaviour of the control laws is started off by presenting the performance obtained for each control law on the reference trajectories used for tuning them.

Results for the roll angle control loop are shown in Figure 4. As can be seen, both controllers track the reference. Moreover, as intended, the response of the system under control is similar for both controllers. There is only a minor discrepancy at around 6s.

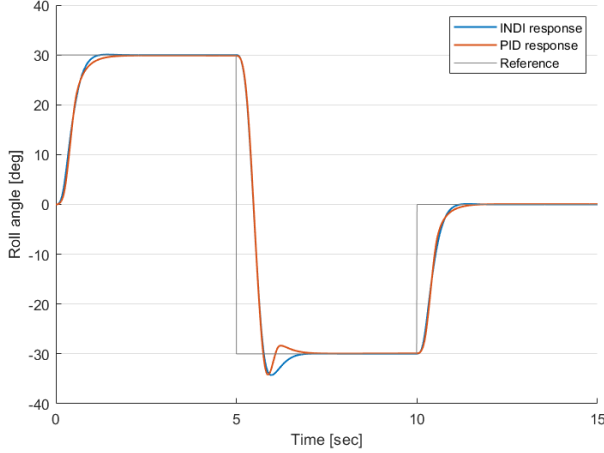


Figure 4: Roll control loop performance on reference trajectory used for tuning

The results for the pitch angle control loop are shown in Figure 5. Again, the response of the system is similar, except for the small discrepancy in the overshoot at 1.5s.

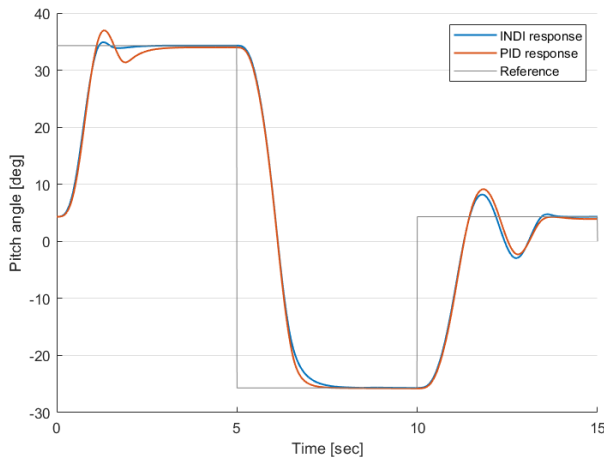


Figure 5: Pitch control loop performance on reference trajectory used for tuning

Lastly, the results for the yaw angle control loop are given in Figure 6. For these trajectories, the general behaviour is similar between both controllers, however, they don't match as closely as for the roll and pitch control loop. Since, in general, the yaw control loop is mainly used to regulate the yaw angle, this is deemed to match sufficiently close for further comparison.

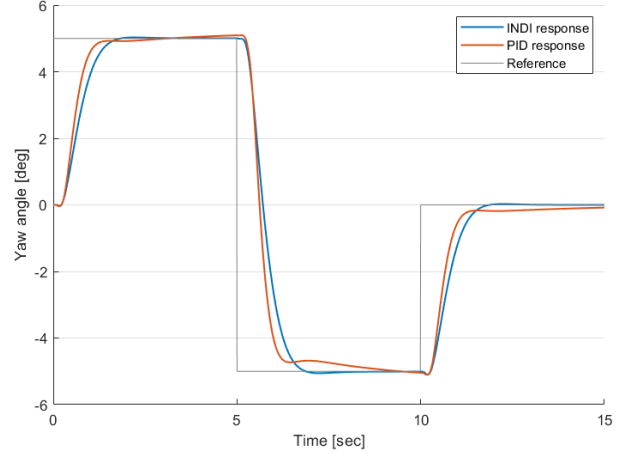


Figure 6: Yaw control loop performance on reference trajectory used for tuning

Table 6 gives a quantitative comparison of the tracking performance for both controllers. The comparison is based on the step response characteristics and root-mean-square error (RMSE). The first RMSE is calculated between the responses of the system, hence should be as small as possible since a zero RMSE indicates that the system has the same response for both controllers. The second RMSE is between the response and the reference, so these should be similar for each control loop, as this indicates that both controllers have a similar tracking error. Note worthy differences are the slightly faster response of the INDI controller for the roll control loop and a higher overshoot of the PID controller for the pitch response as well as a slower settling time.

Table 6: Step response and RMSE characteristics

Characteristic	Roll		Pitch		Yaw	
Step reference [deg]	30		30		5	
RMSE	0.7604		0.7765		0.3883	
	INDI	PID	INDI	PID	INDI	PID
RMSE	23.883	23.627	16.953	16.937	3.857	3.874
Rise time [s]	0.568	0.580	1.005	0.999	0.879	0.628
Settling time [s]	1.012	1.324	1.121	2.696	1.575	1.310
Overshoot [%]	0.261	0	1.708	7.762	0.616	1.874
Peak [deg]	30.080	29.900	30.584	32.664	5.030	5.090
Peak time [s]	1.420	2.700	1.300	1.320	2.200	4.990

For further evaluation of nominal performance, focus is on roll and pitch control since tracking is less relevant for the yaw angle. Evaluation of the controllers is done for a change in dynamic pressure and the tracking of a combined roll and pitch reference.

Change in dynamic pressure

The nominal dynamic pressure conditions are at an altitude of 25000[ft] and velocity of 600[ft/s]. The high dynamic pressure conditions are at 5000[ft] and 900[ft/s]. The results are shown for the roll and pitch reference in Figure 7 and Figure 8, respectively. There is no significant difference between the INDI and PID response, however, the response for both controllers at high dynamic pressure is improved as compared to the nominal dynamic pressure.

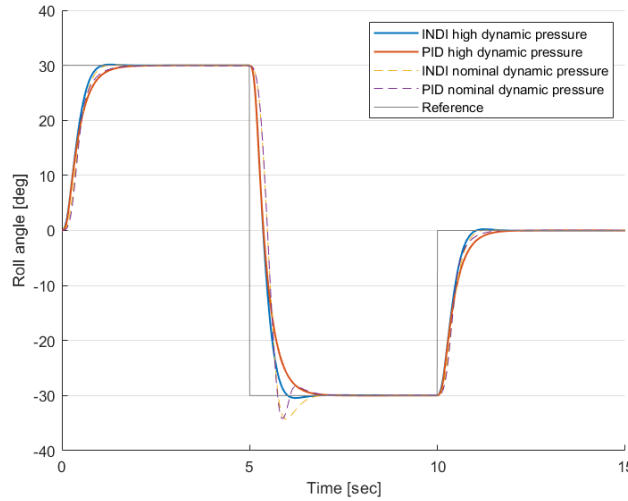


Figure 7: Roll responses for high dynamic pressure

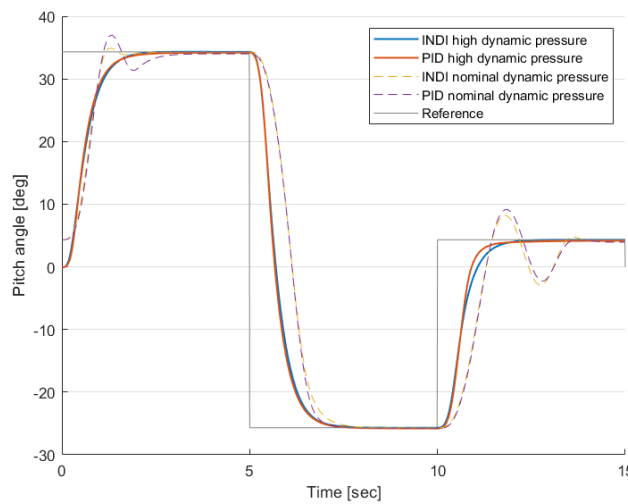


Figure 8: Pitch responses for high dynamic pressure

The low dynamic pressure conditions are at an altitude of 35000[ft] and velocity of 450[ft/s]. Results for the roll and pitch reference are shown in Figure 9 and Figure 10, respectively. The performance for both controllers degrades under these conditions with no significant distinction between the PID and INDI controller.

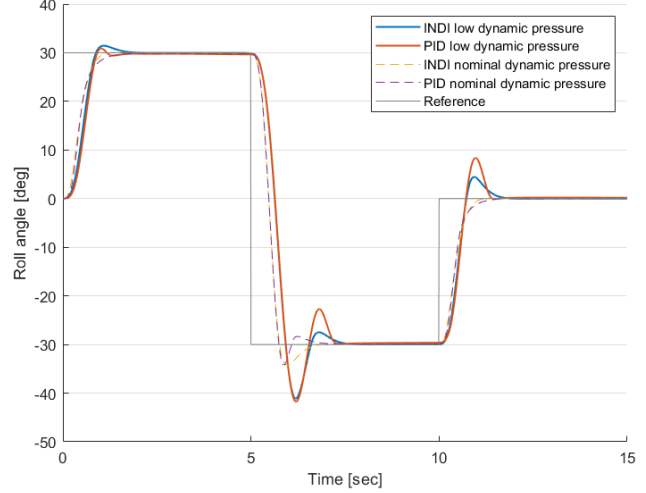


Figure 9: Roll responses for low dynamic pressure

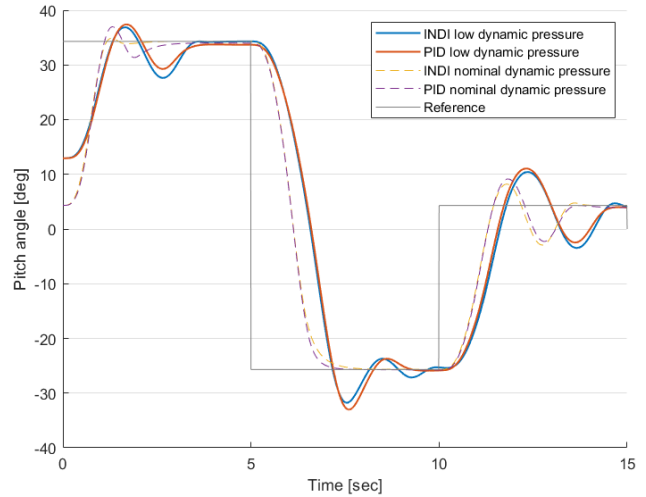


Figure 10: Pitch responses for low dynamic pressure

In terms of performance differences under changing dynamic pressure, INDI performs slightly better in roll angle tracking for higher as well as lower dynamic pressure with a slightly faster rise time and smaller overshoot. Besides this there is no significant difference between the INDI and the PID controller. This is due to the INDI controller still containing two PIDs in the controller architecture, hence the INDI controller still requires gain scheduling for different flight regimes.

Complex maneuver tracking

The complex maneuver consists of a combined roll and pitch maneuver. First, the aircraft pitches up $30[deg]$, then it performs a $360[deg]$ roll, followed by a pitch down of $60[deg]$ and finishing off with a pitch up to its initial pitch angle.

The results for both controllers are shown in Figure 11. The response for both controllers up to 6 seconds at which point the aircraft starts its rolling maneuver. While the INDI controller is capable of maintaining the pitch angle, the response for the PID controller deviates $30[deg]$ from the reference pitch angle. The yaw response, however, deviates by the same range from the reference for both controllers. Furthermore, the sideslip angle reaches a maximum deviation of $\pm 10[deg]$ and maximum angle of attack of $32[deg]$ for both controllers

The INDI controller performs better for this complex maneuver tracking due to it being a nonlinear control law, hence it able to better take into account the coupling between the Euler angles, while the PID controller tries to control each angle separately.

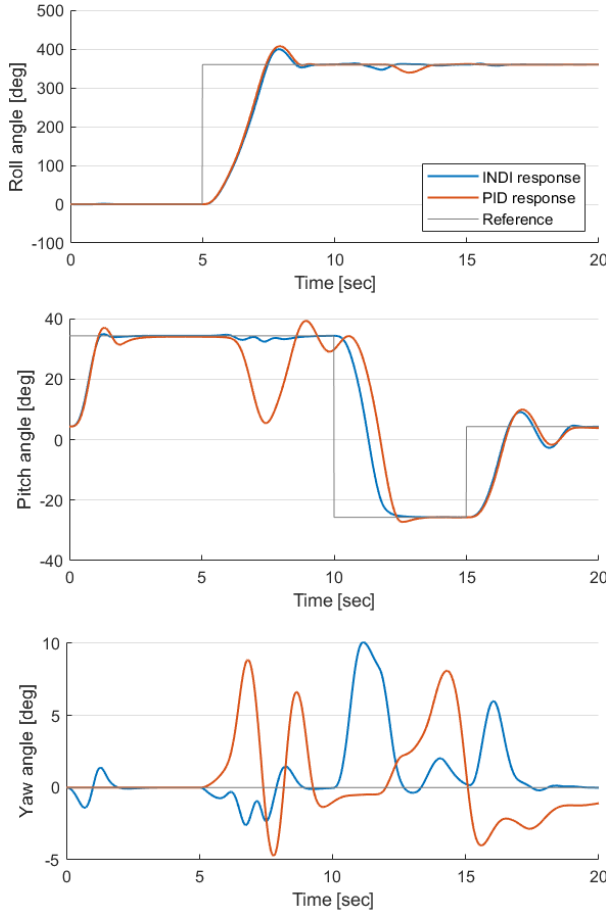


Figure 11: Performance comparison for complex maneuver

B. Analysis Center of Gravity Shift

For previous results, the center of gravity is placed at $0.29\bar{c}$, at which the aircraft is statically stable. Two degrees of static instability are considered in order to evaluate the controller's performance for a relaxed static stability aircraft.

Since the center of gravity position is only shifted longitudinally, only the pitch response is affected, hence the roll response is not included here. Furthermore, the pitch amplitude is set to $6[deg]$ since both controller become unstable for larger pitch maneuvers.

Results are shown in Figure 12 and Figure 13. The major difference are noticed on the pitch down motion, from which it is clear that as the center of gravity shifts aft, the overshoot and oscillations increase. Furthermore, the INDI controller is better able to manage the static instability as it has a lower overshoot and settling time than the PID controller.

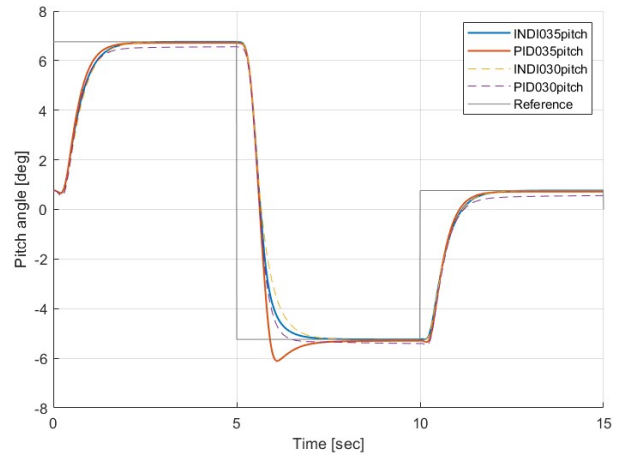


Figure 12: Pitch response for center of gravity location at $0.35\bar{c}$

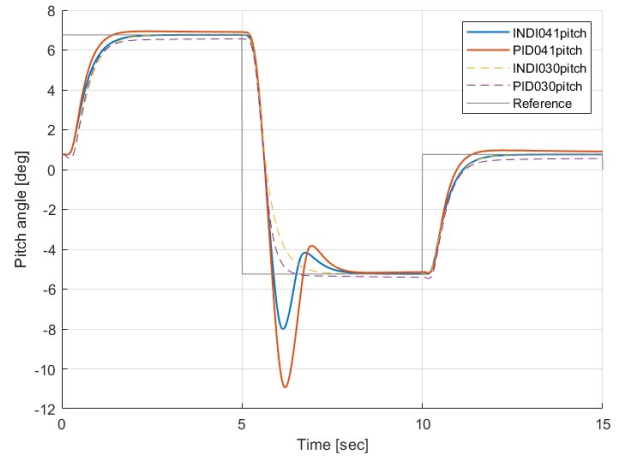


Figure 13: Pitch response for center of gravity location at $0.41\bar{c}$

C. Analysis Model Mismatch

In this section, a comparison of the control laws is made based on their robustness against model mismatch. Specifically, mismatch in the control derivatives and inertia values is evaluated. The evaluation is based on a comparison of the root-mean-square-error (RMSE) between the response and the reference, as explained in subsection C. For each level in deviation from the nominal value of either the control derivatives or the inertia values, a total of 100 simulations are evaluated, each time drawing from the normal distribution. For each simulation run, the RMSE is determined and all results are gathered in boxplots. The evaluation is done based on both a roll doublet reference and pitch doublet reference.

Roll reference

The results for the roll reference are shown in Figure 14. As evident, with increasing levels in deviation of the nominal value of the control derivatives, an increase in spread of the RMSE values can be observed. Interestingly enough, however, the mean does not necessarily increase with an increase in uncertainty. Between PID and INDI, the mean RMSE value for the PID control law is higher for every uncertainty level than for the INDI control law, while the spread, however, is similar between the two.

For the uncertainty in inertia, a similar pattern can be observed. The mean values are higher for the PID control law than for the INDI control law, while the spread is the same. Furthermore, the mean does not shift with increasing level of uncertainty, as highlighted in Table 7, the mean for the PID control law is close to the same at both levels of uncertainty, just like the INDI control law.

These results indicate that INDI-based control law is better in handling both uncertainties in the control derivatives and inertia values. However, with increasing levels of uncertainty in these parameters, the spread in performance also increases equally for both the PID and INDI control laws.

Table 7: Statistical data on model mismatch for a roll reference

		Mismatch in control derivatives					
		0.25		0.15		0.05	
		Control Law		Control Law		Control Law	
RMSE	Upper quartile	INDI	11.575	PID	11.681	INDI	11.587
	Mean		11.350		11.619		11.337
	Lower quartile		10.971		11.102		11.181
		Mismatch in inertia					
		0.10		0.05			
		Control Law		Control Law			
RMSE	Upper quartile		11.540		11.628		11.556
	Mean		10.503		11.594		11.502
	Lower quartile		11.409		11.502		11.419

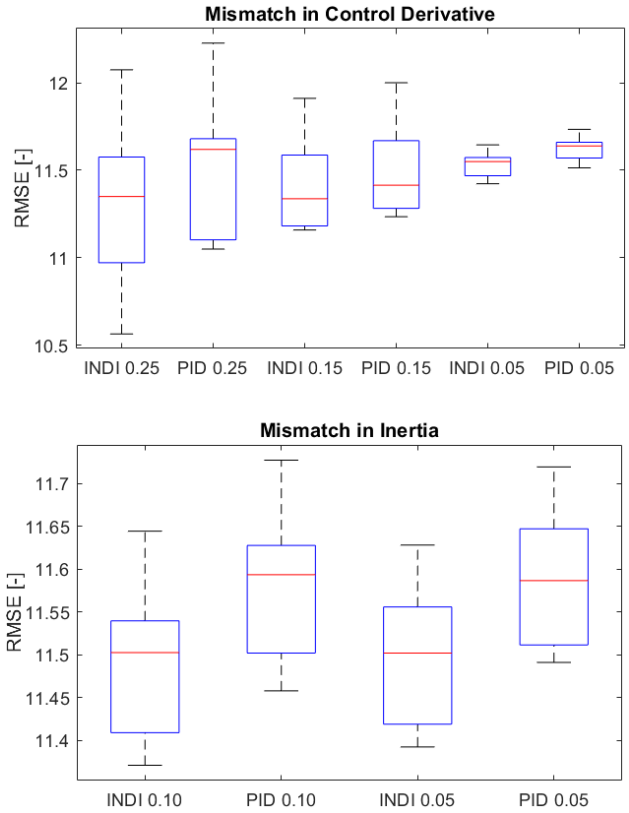


Figure 14: Model mismatch boxplots for roll reference (N=100)

Pitch reference

The results for the pitch reference are shown in Figure 15. From these results, it is clear that in this case, an increase in uncertainty of the control derivatives leads to a higher spread in performance as well as a higher mean value for the RMSE, that is for both the PID and INDI control laws. Additionally, the spread and mean are very similar between the two control laws, as also better indicated by the values given in Table 8.

For the uncertainty in inertia, the spread in performance again increases with an increasing level of uncertainty in the parameters, with a similar spread between both control laws. The mean values, on the other hand, stay relatively constant irrespective of the control law or uncertainty in inertia.

Hence, these results indicate that there is no significant difference between the PID control law and the INDI control law when considering a pitch reference.

Additionally, it is worth noting that the same level of uncertainty, specifically for 0.05, leads to the same spread in performance between a mismatch in control derivatives and a mismatch in inertia. The exact values can be compared from Table 8, where the mean values for the considered cases are highlighted.

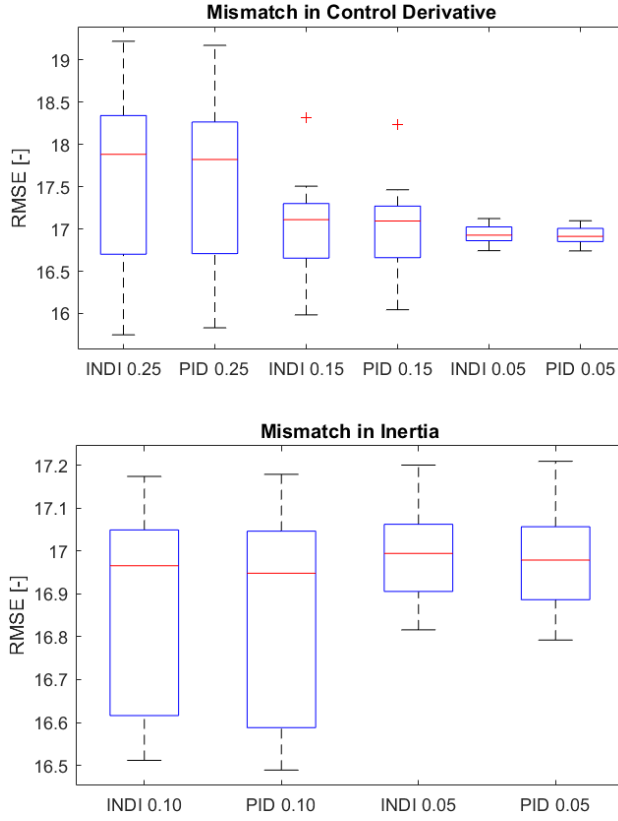


Figure 15: Model mismatch boxplots for pitch reference (N=100)

Table 8: Statistical data on model mismatch for a pitch reference

		Mismatch in control derivatives					
		0.25		0.15		0.05	
RMSE	Control Law	INDI	PID	INDI	PID	INDI	PID
	Upper quartile	18.341	18.266	17.300	17.270	17.024	17.007
	Mean	17.882	17.821	17.111	17.095	16.927	16.912
	Lower quartile	16.702	16.708	16.655	16.660	16.861	16.852
		Mismatch in inertia					
		0.10		0.05			
RMSE	Control Law	INDI	PID	INDI	PID		
	Upper quartile	17.049	17.046	17.062	17.056		
	Mean	16.966	16.948	16.994	16.979		
	Lower quartile	16.616	16.588	16.906	16.886		

D. Analysis System Faults

This section covers the results for the three damage cases introduced in subsection D. For the first damage case, mainly roll performance is affected, for the second damage case, mainly pitch performance and for the third damage case, mainly the yaw performance is affected. Therefore, the damage cases are, respectively, evaluated using a roll, pitch and yaw reference, as used for the analysis of the nominal behaviour of the control laws.

The results for the first damage case are shown

in Figure 16. It can be seen that under the specified system faults, the aircraft is still able to follow the reference roll angle. However, due to the actuators' decreased effectiveness and limited deflection range, the response is slower than the nominal response, which is indicated by the dotted line. This effect is clearly visible at 5 seconds, at which point the reference roll angle changes from $+30[deg]$ to $-30[deg]$. In the nominal case, the aileron is deflected up to $20[deg]$, while in the damaged case, the aileron is saturated at $5[deg]$. This results in the slower response. Additionally, there is also a higher overshoot, caused by the fact that the roll damping has been decreased, and again the decreased actuator effectiveness.

When it comes to the difference between the response of the system under the INDI control law or the PID control law, while the difference is small, the INDI response has a slightly higher rise time at 1 second and a lower overshoot at 6 seconds. The fact that they have a near identical response is also reflected by the aileron deflection angle, which is almost the same for both control laws.

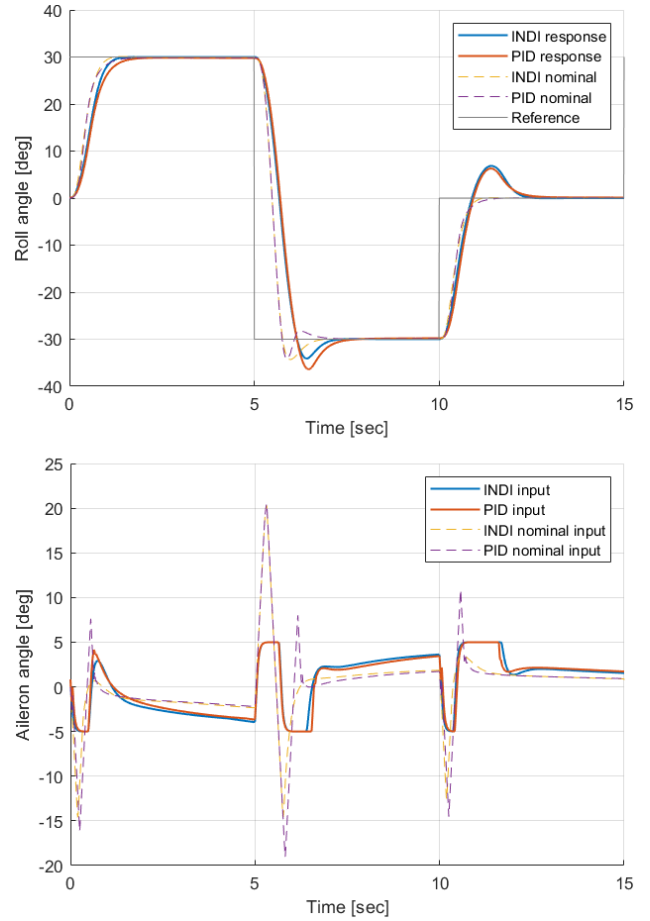


Figure 16: Roll response and aileron inputs for damage case 1

The results for the second damage case are shown in Figure 17. In this damage case, the aircraft fails to track the pitch reference under both control laws. During the first 5 seconds, the elevator is saturated. Regardless, the aircraft is not provided enough pitch up moment to complete the maneuver. The next 5 seconds, the aircraft is able to converge back to the reference, however, slower and with more overshoot than to nominal case. Again, this is due to the elevator being saturated. On the last part of the trajectory, the aircraft is again not capable of converging to the reference due to limited control effectiveness.

For this damage case, again no significant difference can be observed between the INDI response and the PID response. This is expected since the control input is nearly identical. The only discrepancy is at 8 seconds, where the INDI response has slightly less overshoot and a faster convergence to the reference.

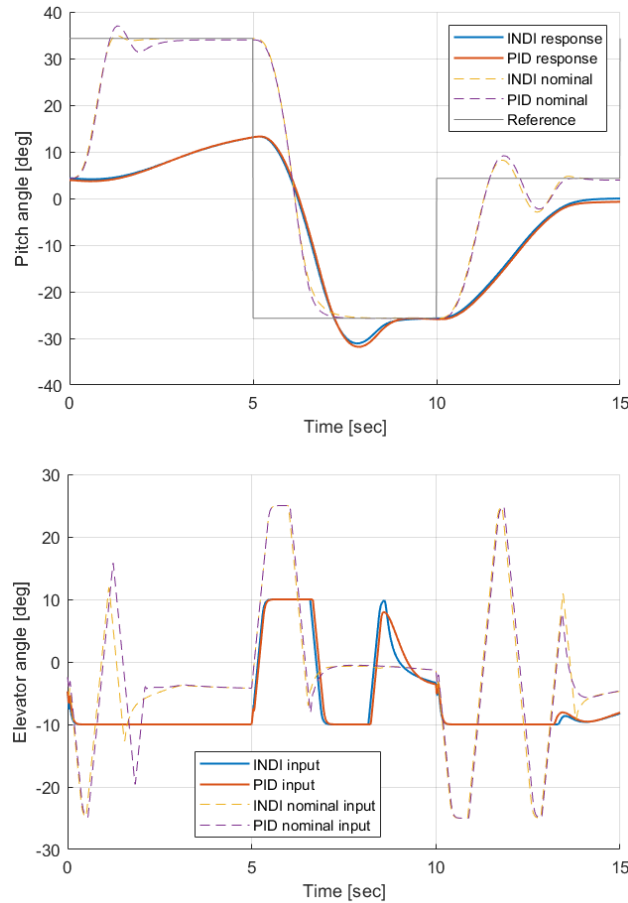


Figure 17: Pitch response and elevator inputs for damage case 2

Lastly, the results for the third damage case are shown in Figure 18. For the first 5 seconds, the actuators are again saturated leading to the aircraft being incapable of tracking the yaw reference. How-

ever, for this response, there is a lot more oscillation, likely due to the reduction in yaw damping. At 5.5 seconds, there is a significant difference between the INDI input and the PID input. The PID-based control law deflects the rudder again to a negative value, leading to a less oscillatory response as compared to the INDI-based control law, which simply decreases the rudder deflection slightly. However, neither control laws are capable of tracking the reference. At 10 seconds, both control laws manage to converge the aircraft back to the reference pitch angle. The INDI control law uses less rudder deflection and has less overshoot in the response.

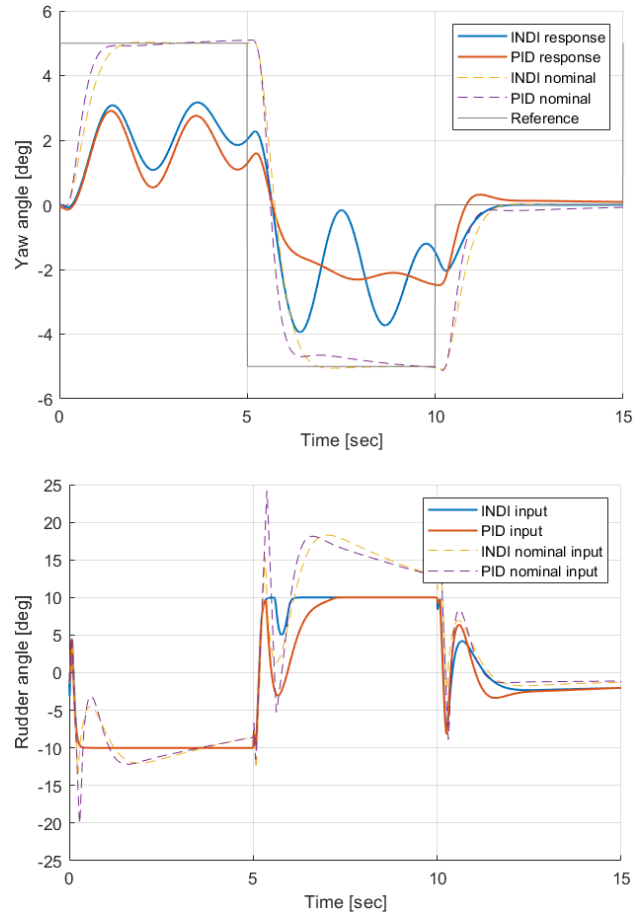


Figure 18: Yaw response and rudder inputs for damage case 3

In general, the INDI- and PID-based control laws have a similar performance under system faults. Both manage to keep the aircraft stable, however, in most cases, performance is significantly affected and limited by the available control authority. In case the control authority is limited too severely, regardless of the control law, the aircraft is incapable of tracking the reference signal.

VII. Conclusions and Recommendations

This paper presented a comparison between Proportional Derivative Integral (PID) control and Incremental Nonlinear Dynamics Inversion (INDI) control applied to the attitude tracking problem of a rigid aircraft.

Both control laws were designed and subsequently tuned as a basis for an equal comparison. The tuning was done through solving an optimization-based problem in which the Root-Mean-Squared Error (RMSE) is minimized for a given pitch, roll and yaw reference. This led to a system response in which the control laws have a comparable performance in the nominal case.

Further analysis focused on tracking of complex references, the effects of changes in dynamic pressure and center of gravity, model mismatch and system defects.

From this analysis, it is shown that INDI does not perform significantly better under changes in dynamic pressure. However, for shifts in center of gravity, INDI manages to control the system better under static instability. In the complex tracking task, INDI also shows a superior performance since it is able to better account for the nonlinear coupling between the attitude angles as compared to the linear PID control law.

In the case of model mismatch, INDI has shown to be more robust against these model uncertainties due to its reduced dependency on accurate knowledge of the system dynamics. Both uncertainty in the control derivatives and the inertia values were considered.

Lastly, three damage cases were considered: damage to the wings, damage to the horizontal stabilizer and damage to the vertical stabilizer. INDI has a slightly better performance than PID control in case of damage to the wings. However, in all damage cases, the control inputs of both control laws are nearly identical, hence the performance is mainly limited by the remaining control effectiveness of the actuators.

Recommendations:

1. In [subsection B](#) and [subsection C](#), high angle-of-attack characteristics of the F-16 aircraft are discussed. Further comparison between both controllers can be done with regards to the high-angle of attack performance of the control laws. Since, at high angles of attack, the coupling becomes more significant, causing the system to behave in a more nonlinear manner. Hence, this could be an interesting point of comparison.

2. Tuning of the control law has a big influence on its performance in certain scenarios. Therefore, a different way of tuning the control laws can be considered as a basis for subsequent comparison. One such method which could be worth exploring is to perform the tuning based on a frequency-domain analysis, tuning the parameters such that a similar frequency response is obtained.
3. In the current test setup, the PID gains are set to be static once tuned. Further analysis could be done with regards to more advanced control architectures, making use of gain scheduling or controlling the sideslip angle instead of the yaw angle.

References

- ¹ Balas, G. J., “Flight Control Law Design: An Industry Perspective,” *European Journal of Control*, Vol. 9, No. 2, 2003, pp. 207–226. <https://doi.org/https://doi.org/10.3166/ejc.9.207-226>, URL <https://www.sciencedirect.com/science/article/pii/S0947358003702763>.
- ² Villaum , F., Fabre, P., Singh, S., and Mora-Camino, F., “Control of Input Redundant Affine Systems: Application to Runway Axis Tracking at Landing,” *IFAC Proceedings Volumes*, Vol. 34, No. 15, 2001, pp. 338–343. [https://doi.org/https://doi.org/10.1016/S1474-6670\(17\)40751-8](https://doi.org/https://doi.org/10.1016/S1474-6670(17)40751-8), URL <https://www.sciencedirect.com/science/article/pii/S1474667017407518>, 15th IFAC Symposium on Automatic Control in Aerospace, Bologna/Forl , Italy, September 2-7, 2001.
- ³ Bauschat, J.-M., “Long-Term Flight-Test Experience: A Nonlinear In-Flight Simulation Approach,” *AIAA Modeling and Simulation Technologies Conference and Exhibit*, 2012. <https://doi.org/10.2514/6.2002-4600>, URL <https://arc.aiaa.org/doi/abs/10.2514/6.2002-4600>.
- ⁴ Brinker, J., and Wise, K., “Flight Testing of Reconfigurable Control Law on the X-36 Tailless Aircraft,” *Journal of Guidance Control and Dynamics*, Vol. 24, 2001, pp. 903–909. <https://doi.org/10.2514/2.4826>.
- ⁵ Sieberling, S., Chu, Q. P., and Mulder, J. A., “Robust Flight Control Using Incremental Nonlinear Dynamic Inversion and Angular Acceleration Prediction,” *Journal of Guidance, Control, and Dynamics*, Vol. 33, No. 6, 2010, pp. 1732–1742. <https://doi.org/10.2514/1.49978>, URL <https://doi.org/10.2514/1.49978>.
- ⁶ Acquatella, P., Falkena, W., Van Kampen, E.-J., and Chu, Q., “Robust Nonlinear Spacecraft Attitude Control using Incremental Nonlinear Dynamic Inversion,” *AIAA Guidance, Navigation, and Control Conference*, 2012. <https://doi.org/https://doi.org/10.2514/6.2012-4623>.
- ⁷ Simplicio, P., Pavel, M., Van Kampen, E.-J., and Chu, Q., “An acceleration measurements-based approach for helicopter nonlinear flight control using Incremental Nonlinear Dynamic Inversion,” *Control Engineering Practice*, Vol. 21, 2013, p. 1065–1077. <https://doi.org/10.1016/j.conengprac.2013.03.009>.
- ⁸ van ’t Veld, R., Van Kampen, E.-J., and Chu, Q., “Stability and Robustness Analysis and Improvements for Incremental Nonlinear Dynamic Inversion Control,” *AIAA SCITECH 2018 Forum*, American Institute of Aeronautics and Astronautics, 2018. <https://doi.org/10.2514/6.2018-1127>.
- ⁹ Smeur, E., Chu, Q., and Croon, G., “Adaptive Incremental Nonlinear Dynamic Inversion for Attitude Control of Micro Air Vehicles,” *Journal of Guidance, Control, and Dynamics*, Vol. 39, 2015, pp. 1–12. <https://doi.org/10.2514/1.G001490>.
- ¹⁰ Smeur, E., Croon, G., and Chu, Q., “Cascaded Incremental Nonlinear Dynamic Inversion Control for MAV Disturbance Rejection,” *Control Engineering Practice*, Vol. 73C, 2018, pp. 79–90. <https://doi.org/10.1016/j.conengprac.2018.01.003>.
- ¹¹ Nguyen, L., Ogburn, M. E., Gilbert, W. P., Kibler, K. S., Brown, P. W., and Deal, P. L., “Simulator study of stall/post-stall characteristics of a fighter airplane with relaxed longitudinal static stability,” Tech. rep., NASA, 09 1979.
- ¹² Alghodhaifi, H., and Lakshmanan, S., “Autonomous Vehicle Evaluation: A Comprehensive Survey on Modeling and Simulation Approaches,” *IEEE Access*, Vol. 9, 2021. <https://doi.org/10.1109/ACCESS.2021.3125620>, URL <https://doi.org/10.1109/ACCESS.2021.3125620>.
- ¹³ van Overeem, S., Wang, X., and Kampen, E.-J. V., “Handling Quality Improvements for the Flying-V Aircraft using Incremental Nonlinear Dynamic Inversion,” *AIAA SCITECH 2023 Forum*, American Institute of Aeronautics and Astronautics, 2023. <https://doi.org/10.2514/6.2023-0105>, URL <https://doi.org/10.2514/6.2023-0105>.